

Spotify Music Analytics

1. Project Overview

This project analyzes Spotify Top-50-world chart data using **27,800 chart records** to understand song popularity, artist performance, chart position, album characteristics, explicit-content patterns, song duration, and changes in popularity over time. The objective is to convert raw chart data into actionable insights for content-performance and music-strategy decisions.

2. Dataset Summary

Metric	Value
Rows / Records	27,800
Columns in raw dataset	11
Distinct Songs	794
Distinct Artists	343
Average Popularity	89.62
Average Song Duration	3.28 minutes
Missing Values	0
Duplicate Records	0

Key Features

- Song, Artist, Popularity and Chart Position
- Duration, Album Type and Total Tracks
- Release Date and Chart Date
- Explicit / Non-Explicit Status
- Album Cover URL for dashboard visualization

3. Exploratory Data Analysis using Python

The analysis began with data loading, structural inspection, descriptive statistics, missing-value checks, duplicate checks, date conversion, feature engineering and KPI calculation. The cleaned data was subsequently saved for PostgreSQL analysis.

4. Data Cleaning & Preparation

- **Data Loading:** Imported the Spotify dataset using pandas.
- **Data Quality Check:** Verified 27,800 records across 11 columns.
- **Missing Data Check:** All columns contain 27,800 non-null values.
- **Duplicate Check:** No duplicate records were detected.
- **Date Standardization:** Converted chart date and release date into datetime format.
- **Feature Engineering:** Created year, month and quarter fields.
- **Duration Conversion:** Converted duration_ms into duration_minutes.
- **Cleaned Dataset:** Saved the prepared data as spotify_cleaned.csv.

5. PostgreSQL Database Integration

The cleaned DataFrame was connected to PostgreSQL and loaded into the **top_50_world** table for structured business analysis. The database verification returned **27,800 records**.

Database Step	Result
Table	top_50_world
Records loaded	27,800
Primary analysis	SQL aggregation, filtering and ranking
Advanced SQL	Subqueries and window functions

6. Analytical Workflow

Python Data Cleaning → Exploratory Data Analysis → PostgreSQL SQL Analysis → DAX Measures → Power BI Dashboard → Business Insights

This workflow demonstrates the end-to-end process expected in a Data Analyst project: data preparation, structured analysis, KPI development, visualization and business interpretation.

7. Data Analysis using SQL (Business Transactions)

1. Top 10 Artists by Chart Records – Identified artists with the largest number of chart observations.

artist	song_count
Taylor Swift	1,871
Billie Eilish	860
Sabrina Carpenter	736
The Weeknd	674
Arctic Monkeys	548
Bad Bunny	537
KAROL G	527
Ariana Grande	487
SZA	484
Travis Scott	473

Insight: Taylor Swift has the highest chart-record presence with 1,871 records, followed by Billie Eilish with 860 and Sabrina Carpenter with 736.

2. Top 10 Most Popular Songs – Retrieved the highest-popularity chart observations.

song	artist	popularity
Die With A Smile	Lady Gaga & Bruno Mars	100
Beautiful Things	Benson Boone	100
Die With A Smile	Lady Gaga & Bruno Mars	100
Beautiful Things	Benson Boone	100
un x100to	Grupo Frontera & Bad Bunny	100
Beautiful Things	Benson Boone	100
Espresso	Sabrina Carpenter	100
i like the way you kiss me	Artemas	100
Espresso	Sabrina Carpenter	100
Die With A Smile	Lady Gaga & Bruno Mars	100

Insight: Multiple chart-date records can belong to the same song, so raw top-10 results should be distinguished from *unique-song* rankings when reporting performance.

3. Top Artists by Average Popularity – Compared artist-level average popularity.

artist	average_popularity
Grupo Frontera & Bad Bunny	99.20
Jung Kook & Latto	98.29
leftovermax	97.50
Yng Lvcas & Peso Pluma	97.33
FIFTY FIFTY	97.27

artist	average_popularity
Lady Gaga & Bruno Mars	96.81
iñigo quintero	95.70
Bizarrap & Peso Pluma	95.60
Mitski	95.52
David Kushner	95.52

8. Album Type & Content Analysis

4. Album Type Distribution – Compared the number of chart records across album, single and compilation types.

album_type	song_count
album	17,261
single	10,496
compilation	43

5. Average Popularity by Album Type – Compared popularity across release types.

album_type	song_count	average_popularity	avg_duration_min
single	10,496	91.86	3.17
album	17,261	88.30	3.35
compilation	43	76.26	3.81

Insight: Singles have the highest average popularity at 91.86, compared with 88.30 for albums and 76.26 for compilations.

6. Explicit vs Non-Explicit Songs – Compared content volume and average popularity.

is_explicit	song_count	average_popularity
False	16,628	89.71
True	11,172	89.49

Insight: Non-explicit records are the larger segment (16,628). Average popularity is very similar between non-explicit (89.71) and explicit (89.49) records.

9. Song Duration & Performance Analysis

7. Average Song Duration – Calculated the average duration after converting milliseconds to minutes.

Metric	Value
Average duration	3.28 minutes
Minimum duration	0.69 minutes
Maximum duration	7.62 minutes

8. Longest Songs – Identified the longest chart observations.

song	artist	duration_minutes
Family Matters	Drake	7.62
Family Matters	Drake	7.62
Family Matters	Drake	7.62
Family Matters	Drake	7.62
Family Matters	Drake	7.62
Family Matters	Drake	7.62

9. Shortest Songs – Identified the shortest chart observations.

song	artist	duration_minutes
Trouble	Eminem	0.69
Saturn Returns Interlude	Ariana Grande	0.70
Saturn Returns Interlude	Ariana Grande	0.70
Saturn Returns Interlude	Ariana Grande	0.70
Saturn Returns Interlude	Ariana Grande	0.70
SMOKE HOUR ★ WILLIE NELSON	Beyoncé	0.85

10. Songs Above Average Popularity – Filtered songs whose popularity exceeds the overall average of 89.62.

song	artist	popularity
Die With A Smile	Lady Gaga & Bruno Mars	100
Die With A Smile	Lady Gaga & Bruno Mars	100
BIRDS OF A FEATHER	Billie Eilish	100
greedy	Tate McRae	100
BIRDS OF A FEATHER	Billie Eilish	100
greedy	Tate McRae	100
Die With A Smile	Lady Gaga & Bruno Mars	100

10. Ranking, Time & Comparative Analysis

11. Artists with More Than 10 Chart Records – Used GROUP BY and HAVING to identify artists with substantial chart presence.

artist	song_count
Taylor Swift	1,871
Billie Eilish	860
Sabrina Carpenter	736
The Weeknd	674
Arctic Monkeys	548
Bad Bunny	537
KAROL G	527
Ariana Grande	487
SZA	484
Travis Scott	473

12. Artist Ranking by Average Popularity – Applied ranking logic to compare artist-level performance.

artist	average_popularity	rank
Grupo Frontera & Bad Bunny	99.20	1
Jung Kook & Latto	98.29	2
leftovermax	97.50	3
Yng Lvcas & Peso Pluma	97.33	4
FIFTY FIFTY	97.27	5
Lady Gaga & Bruno Mars	96.81	6
iñigo quintero	95.70	7
Bizarrap & Peso Pluma	95.60	8
Mitski	95.52	9
David Kushner	95.52	10

13. Distinct Songs by Year – Compared the number of unique songs represented in each year.

year	distinct_songs
2023	425
2024	453

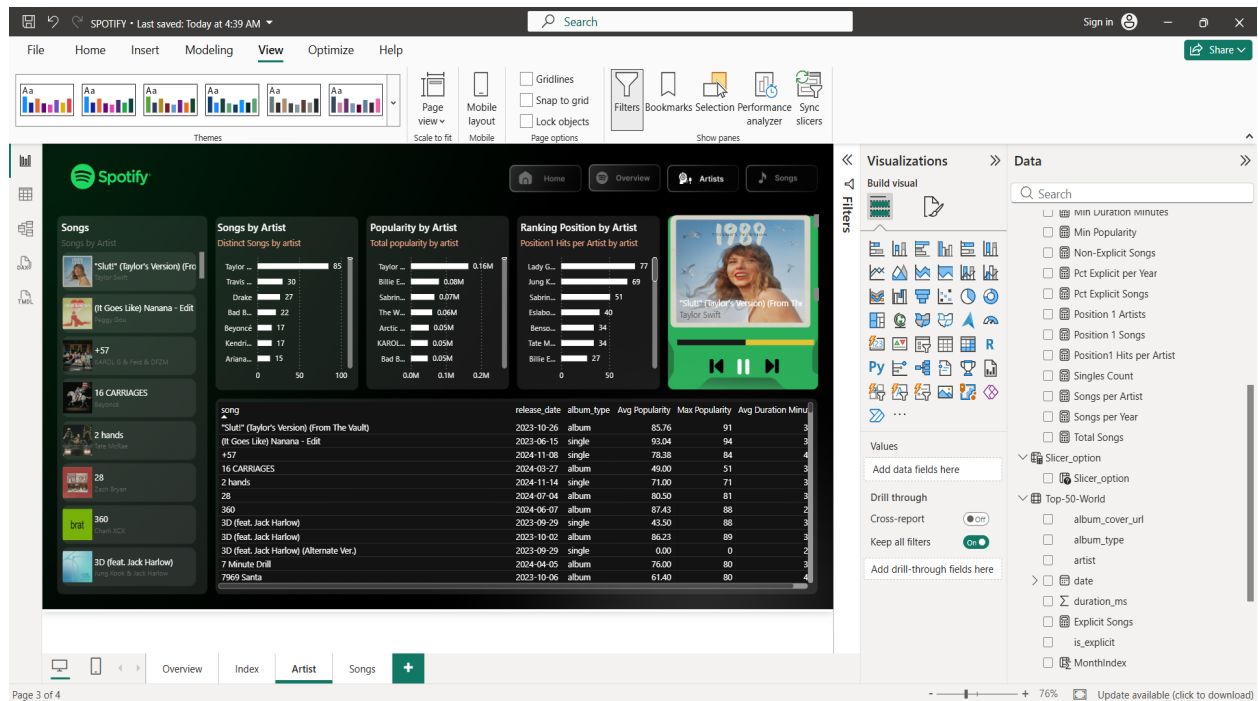
14. Monthly Average Popularity – Examined changes in popularity across months.

month	average_popularity
Jan	92.48
Feb	91.71
Mar	89.81
Apr	89.08

month	average_popularity
May	90.35
Jun	89.24
Jul	89.70
Aug	89.88
Sep	89.88
Oct	86.94
Nov	88.86
Dec	90.45

11. Dashboard in Power BI

Finally, an interactive Power BI dashboard was built to present the analysis visually. The dashboard contains overview KPIs, artist-level analysis, song-level analysis, popularity comparisons, chart-position metrics, album characteristics and detailed performance tables.



Dashboard Area	Purpose
Overview	High-level KPIs and popularity trends
Artist	Artist presence, average popularity and ranking
Songs	Song performance, popularity and detailed records
Interactive filtering	Explore chart performance by available dimensions

Power BI measures include: Total Songs, Distinct Songs, Distinct Artists, Average Popularity, Popularity by Album Type, Explicit/Non-Explicit comparisons, Duration metrics, Position-1 metrics and time-scoped measures.

12. Business Insights

- **Strong artist concentration:** Taylor Swift leads the chart-record count with 1,871 observations.
- **Singles outperform albums on average popularity:** 91.86 vs. 88.30 in this dataset.
- **Content mix:** Non-explicit records represent 16,628 of the 27,800 observations.
- **Popularity is high overall:** the dataset average is 89.62 on the available popularity scale.
- **Song duration:** the average track length is 3.28 minutes, with meaningful variation across songs.
- **Time matters:** monthly average popularity varies, so time-based monitoring adds useful context.
- **Unique-song vs. chart-record distinction:** repeated chart appearances should not be mistaken for unique songs.

13. Business Recommendations

- Use artist-level rankings to identify high-visibility artists for deeper performance analysis.
- Evaluate singles and albums separately when assessing popularity and release performance.
- Monitor monthly popularity trends to identify stronger and weaker periods.
- Use distinct-song metrics when measuring unique content and raw-record metrics when measuring chart presence.
- Combine popularity, chart position, artist, album type, explicit status, duration and time instead of relying on one KPI.
- Use the Power BI dashboard for interactive monitoring and PostgreSQL/Python for deeper investigation.

14. Final Outcome

The project delivers an end-to-end Data Analyst solution: Python for cleaning and EDA, PostgreSQL for structured business analysis, DAX for analytical KPIs, and Power BI for interactive visualization. The final dashboard converts Spotify chart data into a practical view of artist performance, song popularity, release type, content patterns and trends.

Project Stack: Python • Pandas • NumPy • PostgreSQL • SQL • DAX • Power BI